

Irteza Ishaq

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EDUCATION

BS in Artificial Intelligence, FAST NUCES

Aug 2022 – Jun 2026

- **Dean's List** (SGPA: **3.97**) in 7th semester, (SGPA: **3.93**) in 8th semester.
- **Coursework:** Machine Learning/DL, Agentic AI, Database Systems, Recommender Systems, OOP, Data Structures.

SKILLS

Programming Languages: Python, C++, C, Java, Java Scripts

Databases: SQL (PostgreSQL), NoSQL (MongoDB), Supabase

Frameworks & Libraries: FastAPI, React Native, Flutter, Scikit-Learn, Express.js

AI & Data Technologies: Natural Language Processing (NLP), Computer Vision (CV), RAG, LLM Integration, LangChain

DevOps & Cloud: AWS (EC2, S3), CI/CD Pipelines, GitHub Actions, Docker (basic)

Tools & Platforms: n8n, Git, REST APIs, Webhooks

EXPERIENCE

Software Engineer Intern, Folio3 Pakistan

Jun 2025 – Aug 2025

- Achieved **3rd position** in the internship program while mastering the Software Development Life Cycle (SDLC).
- Built a **Health Assistant Application** using **React-Native (Expo)** and **FastAPI**.
- Gained hands-on experience in **Generative AI** and **Agentic AI workflows** through technical sessions.
- Collaborated in an **Agile environment** to understand real-world engineering and deployment pipelines.

Teaching Assistant, Artificial Neural Networks Course, FAST NUCES

Jan 2026 – May 2026

- Assisted the instructor in grading and evaluating **assessments** for **45+ students**.
- Clarified concepts such as **loss functions** and **evaluation metrics**, strengthening understanding of neural network training, optimization, and performance evaluation.

PROJECTS

OmniDocs - Digital Document Management System

GitHub

- Architected a **cloud-enabled** platform for digitizing physical documents into shareable PDFs with secure QR-based access, following **DevOps** and **Lean** development practices.
- Developed backend on **FastAPI**, integrated **AWS S3** for secure storage, and utilized **NeonDB** for efficient data management.
- Containerized the platform using **Docker** and automated deployment pipelines with **GitHub Actions** (CI/CD) on **AWS EC2** to ensure scalable cloud operations.

TPA-Agent - Automated Medical Insurance Claim Orchestration

GitHub

- Built an end-to-end Python pipeline for Gmail-driven medical claims processing, including PDF attachment extraction, email body fallback parsing, and structured clinical field extraction using **Groq LLM** prompts.
- Implemented multi-stage claim validation with local **ICD/CPT/HCPCS** CSV catalogs, **Pinecone** semantic vector search, and policy-document evidence retrieval to determine approved, rejected, or review decisions.
- Designed modular agent architecture with FastAPI orchestration, asynchronous Pub/Sub email triggers, doctor-note classification, insurance limit checking, and optional decision email notification/handoff output.

VectorMind - AI Knowledge Retrieval Platform

GitHub

- Engineered a **RAG-based** PDF Question Answering system using FastAPI, **Pinecone**, and **Hugging Face** embeddings, achieving semantic retrieval confidence scores up to **0.64** on document query.
- Built an automated ingestion pipeline capable of processing **1000+** text chunks with **overlapping chunking (500-token windows)**, vector embedding generation, and metadata indexing for low-latency semantic search.
- Developed **3+ REST APIs** for document ingestion, contextual chat, document listing, and deletion with source attribution, confidence scoring, and hallucination-controlled responses using grounded retrieval techniques.

AI-Powered Carpool Data Pipeline - Event-Driven ETL System

GitHub

- Built a real-time **event-driven ETL pipeline** on **AWS EC2** to extract ride requests from **WhatsApp**, transform them using **Llama 3 (Groq)**, and load structured data into **Supabase**.
- Designed an **LLM-based** validation layer to classify and extract structured entities (time, location, seats) with **98%** accuracy, ensuring high-quality data ingestion and zero noise in the database.
- Implemented an **event-driven notification system** using Supabase listeners to trigger real-time WhatsApp messages via **Green-API** and Flask webhooks.